
Dynamic Evidential Routing: A Neurosymbolic Transformer Interface for Calibrated Clinical Prediction Under Extreme Missingness

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Abstract

1 Clinical predictors can become overconfident when hospital data are sparse, miss-
2 ing, or shifted across sites. We present dynamic evidential routing, a neurosym-
3 bolic transformer interface that maps Monte Carlo dropout uncertainty into Non-
4 Axiomatic Reasoning System (NARS)-style truth values, revises those truth values
5 with explicit clinical rules, and feeds the resulting confidence signal back into atten-
6 tion. The implementation is a NAL truth-value interface rather than a full NARS
7 cognitive architecture. We evaluate TCGA-THCA lymph node metastasis predic-
8 tion as a clinical-genomic proof of concept and WiDS ICU mortality prediction as
9 the primary high-missingness benchmark. On WiDS, the baseline transformer is
10 marginally best on AUC, flat-confidence gating is best on expected calibration error,
11 MC-confidence-only gating is best on accuracy, and the NARS-gated model is best
12 on Brier score. Paired bootstrap intervals for the main within-transformer calibra-
13 tion comparisons include zero, so symbolic revision is not yet statistically isolated
14 as the cause of calibration behavior. The contribution is instead a work-in-progress
15 systems result: the symbolic path is operational, auditable, and active at scale,
16 firing 13,031 times across 8,551 of 13,757 held-out ICU stays while preserving
17 competitive predictive performance.

18 1 Introduction

19 Clinical AI systems can fail by assigning confident probabilities to cases where the evidence is
20 incomplete. Missingness, irregular measurement, and dataset shift are routine in hospital data, and
21 prior work shows that neural predictors can be poorly calibrated under such conditions [Ma et al.,
22 2019, Ovadia et al., 2019, Ghassemi et al., 2021]. A workshop setting is useful for this problem
23 because the key question is not only whether a model ranks patients well, but how communities
24 should expose, contest, and revise uncertainty in high-stakes predictions.

25 Transformers are flexible clinical predictors [Rasmy et al., 2021, Azizi et al., 2021, Li et al., 2019],
26 but attention weights are not evidential confidences. They do not directly say whether a feature
27 was weakly supported, whether a prediction was unstable under epistemic uncertainty, or whether
28 a human-written rule changed the final inference. We therefore study a neurosymbolic interface in
29 which neural predictions and symbolic rules meet in a shared evidential space.

30 Our interface uses NARS truth values (f, c) , where frequency f captures the balance of evidence and
31 confidence c captures evidential support [Wang, 1995, 2006, Wang and Awan, 2013]. The core idea is
32 to map neural uncertainty to an initial truth value, revise that truth value with explicit rules, and feed
33 revised confidence back into attention. This paper makes a narrow claim: dynamic evidential routing
34 is an operational and auditable mechanism for uncertainty-aware clinical inference under missingness,

but the current experiments do not prove that symbolic revision alone improves calibration over simpler confidence gates.

2 Background

Neurosymbolic AI combines neural pattern recognition with symbolic reasoning, knowledge injection, or interpretable rule systems [d’Avila Garcez et al., 2020, Marra et al., 2021, Bhuyan et al., 2024, Colelough and Regli, 2025]. Our method is a tight inference-time interface: the transformer supplies probabilities and attention, while the symbolic layer revises evidential confidence before attention is renormalized.

Uncertainty quantification methods such as Bayesian neural networks, MC dropout, and ensembles are important for clinical AI [Blundell et al., 2015, Gal and Ghahramani, 2016, Lakshminarayanan et al., 2017, Abad et al., 2025], but they do not by themselves provide a symbolic intervention point. Knowledge graphs and symbolic medical reasoning can improve interpretability [Wu et al., 2023, Jiang et al., 2024, Chudasama et al., 2025], but often enrich representations rather than exposing a NARS-style confidence variable that controls neural inference. Dynamic evidential routing sits between these lines of work.

3 Dynamic Evidential Routing

For a statement with positive evidence w^+ and negative evidence w^- , NARS truth values are

$$f = \frac{w^+}{w^+ + w^-}, \quad c = \frac{w^+ + w^-}{w^+ + w^- + k}. \quad (1)$$

We use $k = 1$. Strong deduction combines two premises by

$$f = f_1 f_2, \quad c = f_1 c_1 f_2 c_2, \quad (2)$$

and revision merges two judgments with $\omega_i = c_i / (1 - c_i)$:

$$f = \frac{\omega_1 f_1 + \omega_2 f_2}{\omega_1 + \omega_2}, \quad c = \frac{\omega_1 + \omega_2}{\omega_1 + \omega_2 + 1}. \quad (3)$$

The implementation treats neural and symbolic judgments as distinct sources when revision is invoked, but does not maintain full NARS evidential-base bookkeeping [Wang, 2024].

For a neural probability p and MC-dropout predictive variance σ^2 , the neural initializer is

$$f_{\text{neural}} = p, \quad c_{\text{neural}} = \frac{p(1-p)}{p(1-p) + \sigma^2 + \varepsilon}. \quad (4)$$

This is not a canonical NARS semantics; it is an application-specific bridge from neural uncertainty to NAL-style evidential support. After symbolic revision, attention is updated by

$$\alpha_i^{\text{new}} = \frac{\alpha_i c_i^\gamma}{\sum_j \alpha_j c_j^\gamma}, \quad (5)$$

where γ controls how strongly confidence gates feature attention.

3.1 Inference-time computation

Dynamic evidential routing is intentionally placed at inference time. The transformer is first trained as an ordinary supervised predictor. At test time, repeated MC-dropout passes estimate predictive variance, and the deterministic forward pass provides attention weights. The method then constructs neural truth values, evaluates symbolic rules, revises matching truth values, and renormalizes attention. This separation lets the symbolic component be inspected or edited without retraining the neural model.

The inference path has six stages. First, the trained transformer produces a probability and feature-level attention weights. Second, repeated stochastic passes estimate uncertainty. Third, Eq. 4 initializes neural truth values. Fourth, a proposition extractor evaluates clinical conditions and emits symbolic truth values for rules that fire. Fifth, Eq. 3 merges neural and symbolic truths. Sixth, Eq. 5 produces the final confidence-gated attention distribution.

72 3.2 Trace interface

73 The system exports a trace for every triggered rule. The trace records the feature name, the clinical
74 condition, the neural truth value before revision, the symbolic truth value, the revised truth value, and
75 the attention gain. These traces are not post-hoc natural-language rationales. They are records of the
76 arithmetic that actually changed the prediction path.

77 This trace interface is the core systems contribution. A domain expert can inspect whether the right
78 rule fired, whether the assigned symbolic truth value is clinically defensible, and whether the revised
79 confidence affected attention in the expected direction. If a rule is wrong, the trace localizes the error
80 to a proposition or truth-value assignment. If no rule fires, the system reduces to uncertainty gating.
81 If a rule fires but metrics do not change, the symbolic pathway is still auditable, but the experiment
82 does not support a performance claim.

83 4 Experiments

84 We evaluate two benchmarks with different roles. TCGA-THCA lymph node metastasis prediction
85 is a multimodal proof of concept using 457 labeled cases split into 319 training, 69 validation, and
86 69 held-out test cases. The processed input has 105 dimensions: clinical numeric features, binary
87 mutation indicators, and one-hot clinical categorical features. WiDS Datathon 2020 ICU mortality
88 prediction is the primary scale test, with 91,713 labeled ICU stays split into 64,199 training, 13,757
89 validation, and 13,757 held-out test stays.

90 All transformer variants use $d_{\text{model}} = 64$, four heads, two encoder layers, and 50 MC-dropout passes.
91 We compare a random forest, a baseline transformer, flat-confidence gating with $c = 0.5$, MC-
92 confidence-only gating, and NARS-gated symbolic revision. We report AUC, Brier score, expected
93 calibration error (ECE), and accuracy, with bootstrap intervals for AUC, Brier, and ECE.

94 4.1 Why two benchmarks?

95 TCGA and WiDS answer different questions. TCGA tests whether the same mathematical interface
96 can operate when inputs combine clinical variables, categorical encodings, and a somatic-mutation-
97 derived binary gene matrix. Its held-out split is too small to support strong claims about tiny
98 metric differences, but it is useful for checking whether the interface survives heterogeneous feature
99 construction.

100 WiDS tests the more important high-missingness setting. It contains many more cases, a structured
101 ICU prediction target, and enough held-out examples for calibration summaries to be more meaningful.
102 The paper therefore treats TCGA as a clinical-genomic feasibility check and WiDS as the primary
103 empirical validation.

104 4.2 Baselines and controls

105 The baseline transformer tests whether the neural architecture is competitive before confidence gating
106 is introduced. The flat-confidence control assigns $c = 0.5$ to every feature, isolating the effect of
107 adding a normalized gate without using uncertainty. The MC-confidence-only ablation uses neural
108 uncertainty but no symbolic revision. The NARS-gated model adds symbolic revision on top of the
109 neural confidence initializer.

110 This control ladder is necessary. A comparison between baseline attention and NARS-gated attention
111 alone would not identify the source of any gain. A small improvement could come from attention
112 regularization, neural uncertainty gating, symbolic revision, or noise. The flat-confidence and
113 MC-confidence-only controls make the conclusion more conservative but more credible.

114 4.3 Rule bases

115 The TCGA rule base contains four thyroid-cancer conditions: BRAF mutation, age at least 55
116 years, pathologic T3/T4 stage, and extrathyroid extension. The WiDS rule base contains four ICU
117 conditions: lactate, hypotension, age, and creatinine. These are deliberately small rule bases. They
118 are probes for symbolic intervention, not complete clinical ontologies.

119 The rule bases are applied only at inference time. Each rule has a symbolic truth value and a condition
120 over the processed clinical record. When a condition is satisfied, the symbolic truth value revises the
121 corresponding neural truth value. When no rule fires, the model behaves like the relevant neural or
122 confidence-only baseline.

123 4.4 Metrics

124 AUC measures ranking quality. Brier score measures squared probability error and therefore rewards
125 calibrated probabilities. ECE summarizes reliability-diagram miscalibration using equal-frequency
126 bins. Accuracy is reported at the default threshold for completeness, but it is not the main clinical
127 metric. In high-missingness clinical prediction, calibration and decision-threshold behavior are at
128 least as important as rank ordering.

129 4.5 Preprocessing and leakage controls

130 Both dataset branches use train-split-fitted preprocessing. Continuous variables are imputed and
131 scaled using statistics fit only on the training split, and categorical encodings are fixed before
132 validation and test evaluation. This matters because the paper’s claim concerns calibration under
133 missingness: if imputation or feature selection were allowed to look at the held-out cohort, the
134 resulting confidence estimates would be optimistic in exactly the part of the pipeline that dynamic
135 evidential routing is meant to inspect.

136 The TCGA branch merges structured clinical tables with a mutation-derived binary gene matrix.
137 Mutation files are parsed with comment-aware handling, functionally relevant variant classes are
138 retained, and repeated mutations in the same gene are collapsed to a case-level indicator. Missing
139 mutation indicators are filled with zero after the mutation matrix is joined to the clinical base. The
140 binary nodal-metastasis target is derived from pathologic nodal staging, mapping $N0$ to negative and
141 $N1/N1a/N1b$ to positive while excluding unavailable or indeterminate nodal labels.

142 The WiDS branch uses a fixed ICU feature subset with substantial missingness. This branch is
143 intentionally closer to the expected failure mode for hospital prediction: a model receives a sparse
144 clinical record, preprocessing repairs the matrix enough for supervised learning, and the predictor
145 must still decide how much evidential trust to place in each feature. The NARS interface operates
146 after this preprocessing step rather than replacing it. In other words, imputation provides a usable
147 numeric input, while evidential routing asks whether the resulting neural evidence should be trusted
148 strongly.

149 4.6 Implementation details

150 Each tabular feature is treated as a token-like input to the transformer encoder. The model embeds
151 features into a shared hidden space, applies multi-head self-attention, and aggregates the encoded
152 representation for binary prediction. The dynamic routing layer reads the trained transformer’s
153 feature-level attention weights and the MC-dropout predictive distribution; it does not change the
154 supervised training objective. This design choice keeps the ablation clean because all transformer-
155 family variants share the same underlying predictor and differ only in the inference-time confidence
156 signal used to gate attention.

157 MC dropout is run for 50 stochastic forward passes. The predictive mean supplies the probability
158 used as f_{neural} , and the predictive variance supplies the uncertainty term in Eq. 4. The implementation
159 uses a small numerical ε so that near-deterministic predictions do not create unstable divisions. The
160 attention update in Eq. 5 is then applied after rule revision. Unless otherwise stated, we use $\gamma = 2$ as
161 the default confidence-gating exponent.

162 The exported artifacts are part of the method rather than only a convenience for reproduction. For
163 each benchmark, the pipeline writes metric summaries, bootstrap intervals, rule-firing counts, and
164 case-level traces. The anonymous repository exposes the same entry point for both branches, allowing
165 reviewers to inspect whether a metric was produced with the intended split, whether a rule fired on a
166 held-out case, and whether the revised confidence changed attention in the expected direction. The
167 code path therefore supports the paper’s auditability claim directly.

Table 1: Worked simulation trace for neural-symbolic revision and confidence-gated attention.

Quantity	Value	Source
Neural truth	(0.8000, 0.7619)	Eq. 4, $p = 0.8$, $\sigma^2 = 0.05$
Symbolic truth	(0.9000, 0.7000)	Given rule
Revision weights	(3.2000, 2.3333)	$\omega = c/(1 - c)$
Revised truth	(0.8422, 0.8469)	Eq. 3
Gain	0.7173	c_{rev}^2

Table 2: Held-out TCGA-THCA performance for lymph node metastasis prediction.

Variant	AUC	Brier	ECE	Acc.
Random forest	0.6613 [0.5101, 0.7908]	0.2200 [0.1895, 0.2498]	0.1597 [0.1317, 0.2824]	0.5942
Baseline transformer	0.7252 [0.5931, 0.8454]	0.2118 [0.1800, 0.2449]	0.1390 [0.1183, 0.2610]	0.6667
Flat-confidence	0.7261 [0.5934, 0.8468]	0.2118 [0.1793, 0.2455]	0.1390 [0.1163, 0.2597]	0.6812
MC-confidence-only	0.7261 [0.5957, 0.8460]	0.2122 [0.1808, 0.2451]	0.1403 [0.1177, 0.2621]	0.6667
NARS-gated	0.7261 [0.5957, 0.8460]	0.2121 [0.1805, 0.2452]	0.1400 [0.1176, 0.2618]	0.6667

5 Results and Discussion

On TCGA-THCA, flat-confidence, MC-confidence-only, and NARS-gated transformers tie for best default AUC at 0.7261, while flat-confidence has the best Brier score, ECE, and accuracy. Because the test set contains only 69 cases and the confidence intervals overlap heavily, this result should be read as a multimodal feasibility result, not as a superiority claim.

On WiDS, the transformer family is again tightly clustered. The baseline is marginally best on AUC, flat-confidence is best on ECE, MC-confidence-only is best on accuracy, and NARS-gated is best on Brier score. The NARS-vs-baseline paired bootstrap intervals include zero for Brier and ECE, and so do the NARS-vs-flat-confidence intervals. Thus the current run does not isolate the symbolic rule path from neural uncertainty gating alone.

The useful systems signal is that symbolic rules fired on 8,551 of 13,757 held-out ICU stays, producing 13,031 feature-level revisions. The pathway is therefore physically active and auditable, even though its marginal effect over simpler confidence gates remains unresolved. This motivates workshop discussion around stronger rule bases, external validation, and better protocols for distinguishing uncertainty gating from symbolic intervention.

5.1 Rule activity

On TCGA, symbolic rules fired 79 times across 42 of 69 held-out cases. Per-rule counts were 0 for BRAF mutation, 25 for age at least 55 years, 29 for pathologic T3/T4 stage, and 25 for extrathyroid extension. This result is primarily a feasibility signal: the symbolic pathway remained connected when the model used clinical and genomic inputs.

On WiDS, rule activity was much larger. ICU rules fired on 8,551 of 13,757 held-out stays. The per-rule counts were 1,936 for lactate, 5,338 for hypotension, 3,443 for age, and 2,314 for creatinine. The rule pathway therefore operated at hospital-cohort scale rather than only on isolated examples.

5.2 Interpreting the controls

The control ladder is the most important part of the empirical design. A plain transformer answers whether the neural architecture is a credible baseline. Flat-confidence attention answers whether merely inserting a normalized attention gate changes calibration. MC-confidence-only attention answers whether neural uncertainty is already sufficient. NARS-gated attention answers whether adding symbolic revision on top of uncertainty changes behavior further.

The current results place most of the visible improvement inside the broad family of confidence-gated attention rather than uniquely inside symbolic revision. This is especially clear on WiDS, where flat-confidence, MC-confidence-only, and NARS-gated variants are numerically very close. That does not make the symbolic layer irrelevant. It means the next experiment must be designed to isolate

Table 3: Held-out WiDS ICU performance for hospital mortality prediction.

Variant	AUC	Brier	ECE	Acc.
Random forest	0.8754 [0.8655, 0.8858]	0.05821 [0.05524, 0.06149]	0.00764 [0.00547, 0.01242]	0.9259
Baseline transformer	0.8829 [0.8732, 0.8926]	0.05621 [0.05334, 0.05946]	0.00601 [0.00466, 0.01050]	0.9288
Flat-confidence	0.8829 [0.8731, 0.8925]	0.05618 [0.05328, 0.05946]	0.00490 [0.00411, 0.00969]	0.9288
MC-confidence-only	0.8829 [0.8731, 0.8925]	0.05618 [0.05327, 0.05945]	0.00495 [0.00386, 0.00955]	0.9291
NARS-gated	0.8829 [0.8731, 0.8925]	0.05618 [0.05327, 0.05945]	0.00494 [0.00387, 0.00956]	0.9288

symbolic content more aggressively, for example by increasing rule coverage, pre-registering rule truth values, and testing cases where clinical rules are expected to contradict the learned association.

This distinction is central for workshop discussion. A neurosymbolic system should not be credited merely because a symbolic module is present. It should be credited when the symbolic module changes a measurable failure mode, exposes a useful reasoning path, or improves human correction. The present paper supports the second of these claims most strongly: the symbolic path is active and traceable. The first claim, a statistically isolated calibration improvement, remains open.

5.3 Calibration interpretation

The WiDS calibration result should be read carefully. The NARS-gated variant has the best Brier score at the reported precision, but flat-confidence has the best ECE, and MC-confidence-only is nearly indistinguishable from NARS-gated attention. Paired bootstrap intervals for NARS-vs-baseline Brier and ECE include zero, and so do the paired intervals for NARS-vs-flat-confidence.

This means the current evidence does not show that symbolic revision caused a statistically separable calibration gain. That restraint is important for a workshop paper: the interesting contribution is not an exaggerated leaderboard claim, but a system design that makes symbolic intervention part of the computation and exposes where it acted.

5.4 Gamma sensitivity

The TCGA gamma sweep over $\gamma \in \{0.25, 0.5, 1, 2, 4\}$ shows a mild ranking-versus-calibration trade-off. Stronger gating increases NARS-gated AUC slightly, peaking at 0.7294 when $\gamma = 4$, while Brier score worsens from 0.21186 to 0.21244. On such a small split, this is sensitivity analysis, not evidence for tuning a superior regime.

The practical lesson is that attention confidence should not be tuned only for AUC. If the purpose is clinical decision support under uncertainty, a stronger gate may change ranking, calibration, and threshold behavior in different directions. Future work should therefore pre-specify which decision metric the gate is meant to optimize.

5.5 Decision-curve interpretation

Decision-curve analysis over threshold probabilities from 0.05 to 0.95 shows that transformer variants are broadly similar, with modest threshold-dependent differences at intermediate thresholds. This is consistent with the main metric tables: the interface does not create a uniform performance improvement across all operating points.

For clinical triage, threshold-dependent behavior matters. A model that is indistinguishable by AUC may still change decisions near a treatment or escalation threshold. The trace interface is valuable in that setting because it can show whether a symbolic rule participated in a probability shift near the decision boundary.

5.6 Trace semantics

The trace should be read as a computational audit trail rather than an explanation generated after the fact. For each fired rule, the trace records the original neural truth value, the symbolic truth value, the revised truth value, and the attention gain implied by the revised confidence. This gives the reviewer a path from a human-readable proposition to a numerical change in the model’s computation.

240 There are two useful trace-level questions. The first is factual: did the rule fire on the right case and
241 the right feature? This tests proposition extraction and feature mapping. The second is normative:
242 was the symbolic truth value reasonable for the clinical context? This tests rule curation. The metric
243 tables alone cannot answer either question, which is why the trace interface is necessary for the
244 neurosymbolic claim.

245 5.7 What would isolate symbolic revision?

246 The current ablations show that the symbolic path is active, but not that it is uniquely responsible for
247 calibration behavior. A stronger isolation study would include three locked components. First, the
248 uncertainty estimator should be fixed before rule evaluation. Second, the rule base and all symbolic
249 truth values should be frozen before test evaluation. Third, the analysis should predefine a subset
250 of cases where symbolic evidence is expected to matter, such as high-risk physiologic patterns with
251 high neural variance.

252 The strongest evidence would be a paired comparison where NARS-gated attention improves Brier
253 score, ECE, or decision utility over baseline, flat-confidence, and MC-confidence-only controls
254 with intervals excluding zero. A complementary audit result would show that the improved cases
255 correspond to clinically plausible rule firings rather than accidental regularization. Until such an
256 experiment is complete, the paper’s claim remains deliberately modest.

257 5.8 Relation to missingness

258 Missingness creates two coupled problems. The first is representation: the model must encode
259 incomplete records. The second is trust: the model must avoid treating repaired or weakly observed
260 records as if they were fully supported. Most tabular clinical pipelines address the first problem
261 through imputation, missingness indicators, or architectures robust to sparse inputs. Dynamic
262 evidential routing addresses the second problem by making confidence an explicit intermediate
263 variable.

264 The WiDS benchmark is therefore not simply a larger dataset. It is a stress test for the specific
265 reason this interface exists. ICU variables can be missing because they were not measured, because
266 they were clinically unnecessary, because measurement happened outside the observed window, or
267 because workflows differ across sites. These mechanisms can carry information, but they can also
268 create brittle shortcuts. A symbolic rule that fires on a physiologic condition should be evaluated
269 together with the confidence assigned to the underlying feature evidence.

270 5.9 Decision utility and clinical use

271 Clinical prediction is rarely consumed as a raw ranking. A risk score is usually mapped to a
272 thresholded action, a triage priority, or a request for additional review. That makes the distinction
273 between AUC, calibration, and net benefit practically important. A model with nearly unchanged
274 AUC can still alter how many patients cross a clinically meaningful threshold.

275 Dynamic evidential routing is most naturally used as a review tool at such thresholds. When a
276 prediction changes near an operating point, the trace can show whether the change followed from
277 high epistemic uncertainty, from a fired symbolic rule, or from a combination of both. This is not a
278 substitute for prospective validation, but it is a more inspectable failure-analysis surface than a scalar
279 probability alone.

280 6 Additional Analysis

281 The result pattern suggests a conservative interpretation of dynamic evidential routing as an instru-
282 mentation layer first and a performance-improvement layer second. On TCGA, the small held-out
283 split makes all transformer-family differences unstable. On WiDS, the larger held-out split makes the
284 calibration summaries more informative, but the strongest conclusion is still that multiple confidence-
285 gated variants are close.

286 6.1 Why Brier and ECE can disagree

287 Brier score and ECE measure related but different behavior. Brier score is a smooth proper scoring
288 rule over individual probabilities. ECE bins predictions and compares average confidence to empirical
289 frequency inside each bin. A model can slightly improve Brier score while leaving binned reliability
290 essentially unchanged, or can improve binned reliability while not improving squared probability
291 error. This explains why the WiDS table should not be reduced to a single winner.

292 For a clinical workshop audience, this disagreement is useful. It forces the evaluation to specify what
293 kind of reliability matters. If the downstream use is thresholded triage, local calibration around the
294 operating threshold may matter more than global ECE. If the downstream use is risk communication,
295 probability-level scoring may be more relevant. Future versions should therefore report reliability
296 diagrams, threshold-local calibration, and decision curves together rather than treating any single
297 scalar as definitive.

298 6.2 Why the rule base is intentionally small

299 The rule bases are small because this paper is testing an interface, not claiming that four rules
300 constitute a clinical ontology. A broad rule base would make the demonstration look more realistic,
301 but it would also make debugging harder: if a metric changed, one would not know whether the
302 change came from the interface, a particular rule family, or accidental interactions among many rules.

303 The four-rule design gives a readable first test. Each rule is easy to inspect, the firing counts are
304 easy to report, and the trace for a single case can be checked by hand. Once this narrow interface is
305 validated, the natural next step is to replace the small probe rule base with a curated clinical rule set
306 and evaluate whether rule coverage, rule conflict, and truth-value uncertainty affect calibration.

307 6.3 Uncertainty estimator sensitivity

308 MC dropout is a pragmatic uncertainty estimator, but it is not the only candidate. Deep ensembles
309 may produce better uncertainty under dataset shift, at greater computational cost. Bayesian neural
310 networks provide a more direct probabilistic treatment of parameters but can be difficult to scale and
311 tune. Conformal methods can provide coverage-style guarantees, but they do not immediately give
312 the feature-level confidence quantity needed by Eq. 5.

313 This creates an important design axis for neurosymbolic systems. If the uncertainty estimator is weak,
314 the symbolic interface may receive overconfident neural truth values and revise them insufficiently. If
315 the estimator is too conservative, the confidence gate may attenuate many features indiscriminately
316 and behave like a global regularizer. The paper’s gamma sensitivity results are an early sign of this
317 trade-off, but a full uncertainty-estimator comparison remains future work.

318 6.4 Clinical rule governance

319 Symbolic truth values are not self-justifying. In a clinical setting, a rule such as elevated lactate
320 should have a documented source, an intended population, and a confidence assignment that can
321 be reviewed. If two experts disagree on the confidence of a rule, the system should either report
322 a sensitivity interval over plausible truth values or maintain alternative rule versions. The present
323 implementation does not solve this governance problem, but it exposes where such governance would
324 attach.

325 This is another reason the work belongs in a workshop setting. The right standard is not only whether
326 a symbolic rule improves a benchmark score. It is whether the community can define processes for
327 curating, contesting, and updating such rules without turning the model into an opaque collection of
328 hand-tuned exceptions. Dynamic evidential routing supplies one concrete interface through which
329 that discussion can be made technical.

330 6.5 External validation criteria

331 Before any clinical claim, the model should be evaluated on an institutionally or temporally inde-
332 pendent cohort with frozen preprocessing, frozen neural weights, frozen rule definitions, and frozen
333 truth-value assignments. The evaluation should report the same four controls used here. It should

also report the number of rule firings, the proportion of cases where attention changes materially, and the distribution of probability shifts near clinically relevant thresholds.

An external study should define success in advance. A reasonable target would be improved calibration or decision utility over all transformer-family controls, accompanied by trace audits showing clinically plausible rule participation in changed cases. If the NARS-gated model merely matches the MC-confidence-only model, the conclusion should remain that symbolic routing is an auditability feature, not a proven calibration improvement.

7 Workshop Relevance

This paper is well suited to a NeurIPS workshop because it raises unresolved questions rather than closing the topic. The result suggests that symbolic intervention can be made active and auditable inside a neural prediction pipeline, but it also shows that ordinary confidence gating can explain much of the observed metric behavior. The community question is how to design evaluations that distinguish these effects.

The first open question is how to initialize evidential confidence. MC dropout is inexpensive, but it can underestimate uncertainty under shift. Deep ensembles may be more reliable but are more expensive. Conformal methods may offer coverage guarantees, but they do not directly produce NARS-style evidential support. A workshop discussion could clarify which uncertainty signals are appropriate for neurosymbolic revision.

The second open question is how to curate symbolic truth values. Clinical rules are rarely binary certainties. Their strength may depend on cohort, measurement timing, comorbidities, and missingness mechanism. A useful rule base should therefore be reviewed by multiple domain experts and tested with sensitivity analysis over (f, c) assignments.

The third open question is how to evaluate auditability. Standard metrics measure predictive behavior, but not whether a clinician can inspect and correct a reasoning path. Dynamic evidential routing creates traces that can be reviewed, but the paper does not yet include human-in-the-loop studies. Such studies would be natural workshop follow-ups.

8 Limitations and Open Questions

The current experiments use internal held-out splits, not external clinical validation cohorts. The TCGA test set is small, the rule bases are hand-written and intentionally minimal, and the MC-dropout initializer can inherit miscalibration under shift. The implementation also uses NAL truth-value functions without the broader memory and control machinery of full NARS.

The main open questions are practical and communal: Which uncertainty estimators are reliable enough to initialize evidential confidence? How should clinical rule truth values be curated and audited? What benchmark designs can separate generic confidence gating from symbolic revision? These questions make the work especially suited to a NeurIPS workshop setting.

8.1 Failure modes

The interface can fail if uncertainty is misestimated, if proposition extraction is misaligned with encoded features, or if symbolic truth values are clinically wrong. These are not cosmetic issues. A bad confidence estimate can over-weight weak neural evidence; a bad extractor can fire a rule on the wrong feature; a bad rule can make the symbolic pathway confidently wrong.

The benefit of the design is that these failures are inspectable. A trace can reveal whether a rule fired, what truth value it supplied, and how attention changed. That does not make the model safe by itself, but it gives researchers a concrete debugging surface that pure neural models usually lack.

8.2 External validation plan

The next validation stage should freeze preprocessing, transformer weights, rule definitions, and truth values before applying the model to an external cohort. The evaluation should compare baseline attention, flat-confidence attention, MC-confidence attention, and NARS-revised attention. Symbolic

revision would be empirically isolated only if the NARS-revised variant improves calibration or decision utility beyond all three controls with paired intervals that exclude zero.

The same external protocol should report trace quality. For example, reviewers could sample cases where symbolic rules changed attention and ask whether the rule activation is clinically plausible. This would connect quantitative metrics with the auditability claim.

9 Conclusion

Dynamic evidential routing turns neural uncertainty into an explicit NARS-style confidence signal, revises it with symbolic rules, and uses it to control transformer attention. The current evidence supports feasibility, auditability, and scale, not a broad claim of symbolic-gating superiority. Future work should evaluate stronger uncertainty estimators, larger clinical rule bases, external cohorts, and targeted ablations that isolate the contribution of symbolic revision.

Ethics, Data, and Code Availability

This work uses public, de-identified benchmark data and did not collect new patient data. The anonymous repository for review is available at <https://anonymous.4open.science/r/NGTA/>. Large language models were used to assist with drafting, mathematical formulation, and code generation; the author(s) independently checked the derivations, implementation, and reported results and take full responsibility for the work.

References

- A. Abad et al. Uncertainty quantification for machine learning in healthcare: A survey. *arXiv:2505.02874*, 2025.
- S. Azizi et al. Big self-supervised models advance medical image classification. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 3478–3488, 2021.
- B. P. Bhuyan et al. Neuro-symbolic artificial intelligence: A survey. *Neural Computing and Applications*, 36(12):6397–6423, 2024.
- C. Blundell et al. Weight uncertainty in neural networks. In *Proceedings of the International Conference on Machine Learning*, pages 1613–1622, 2015.
- Y. Chudasama et al. Towards interpretable hybrid AI: Integrating knowledge graphs and symbolic reasoning in medicine. *IEEE Access*, 13:12345–12360, 2025.
- B. C. Colelough and W. Regli. Neuro-symbolic AI in 2024: A systematic review. *arXiv:2501.05435*, 2025.
- A. d’Avila Garcez et al. Neurosymbolic AI: The 3rd wave. *Artificial Intelligence*, 294:103504, 2020.
- Y. Gal and Z. Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the International Conference on Machine Learning*, pages 1050–1059, 2016.
- M. Ghassemi et al. Review of challenges and opportunities for machine learning in healthcare. *BMJ*, 372:n4, 2021.
- P. Jiang et al. Reasoning-enhanced healthcare predictions with knowledge graph community retrieval. *arXiv:2410.04585*, 2024.
- B. Lakshminarayanan et al. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- F. Li et al. Fine-tuning bidirectional encoder representations from transformers (BERT)-based models on large-scale electronic health record notes: An empirical study. *JMIR Medical Informatics*, 7(3):e14830, 2019.
- J. Ma et al. Improving rare disease classification using imperfect knowledge graph. *BMC Medical Genomics*, 12(Suppl. 10):195, 2019.
- G. Marra et al. Neural-symbolic computing: An effective methodology for principled integration of machine learning and reasoning. *Journal of Applied Logic*, 6, 2021.

424 Y. Ovadia et al. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift. In
425 *Advances in Neural Information Processing Systems*, volume 32, 2019.

426 L. Rasmy et al. Med-BERT: Pretrained contextualized embeddings on large-scale structured electronic health
427 records for disease prediction. *npj Digital Medicine*, 4(1):86, 2021.

428 P. Wang. *Non-Axiomatic Reasoning System: Exploring the Essence of Intelligence*. PhD thesis, Indiana
429 University, 1995.

430 P. Wang. *Rigid Flexibility: The Logic of Intelligence*. Springer, 2006.

431 P. Wang. Non-axiomatic logic: A model of intelligent reasoning. Second edition draft manuscript dated
432 December 14, 2024, shared by author, 2024.

433 P. Wang and S. Awan. Reasoning in non-axiomatic logic: A case study in medical diagnosis. In *Proceedings of*
434 *the Conference on Artificial General Intelligence*, pages 185–190, 2013.

435 X. Wu et al. Medical knowledge graph: Data sources, construction, reasoning, and applications. *Big Data*
436 *Mining and Analytics*, 6(2):201–217, 2023.

437 A NeurIPS Checklist

438 NeurIPS Paper Checklist

439 1. Claims

440 Question: Do the main claims made in the abstract and introduction accurately reflect the
441 paper’s contributions and scope?

442 Answer: [\[Yes\]](#).

443 Justification: The abstract and Section 1 explicitly state the narrow claim: the interface
444 is operational, auditable, and active at scale, but symbolic revision is not yet statistically
445 isolated as a within-family calibration gain.

446 2. Limitations

447 Question: Does the paper discuss the limitations of the work performed by the authors?

448 Answer: [\[Yes\]](#).

449 Justification: Section 8 discusses internal validation only, small TCGA sample size, hand-
450 written rule bases, possible MC-dropout miscalibration, and the fact that this is not a full
451 NARS implementation.

452 3. Theory assumptions and proofs

453 Question: For each theoretical result, does the paper provide the full set of assumptions and
454 a complete (and correct) proof?

455 Answer: [\[N/A\]](#).

456 Justification: The paper defines an interface and reports empirical results, but it does not
457 claim new theorems. The equations and assumptions used by the method are stated in
458 Section 3.

459 4. Experimental result reproducibility

460 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
461 perimental results of the paper to the extent that it affects the main claims and/or conclusions
462 of the paper (regardless of whether the code and data are provided or not)?

463 Answer: [\[Yes\]](#).

464 Justification: Section 4 reports the datasets, split sizes, model family, baselines, MC-dropout
465 passes, and metrics. The anonymous repository is linked in the data and code statement.

466 5. Open access to data and code

467 Question: Does the paper provide open access to the data and code, with sufficient instruc-
468 tions to faithfully reproduce the main experimental results, as described in supplemental
469 material?

470 Answer: [Yes].

471 Justification: The anonymous repository is available at <https://anonymous.4open.science/r/NGTA/>. The benchmarks are public datasets, and the repository contains

472 reproduction materials for review.

473

474 **6. Experimental setting/details**

475 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-

476 rameters, how they were chosen, type of optimizer) necessary to understand the results?

477 Answer: [Yes].

478 Justification: Section 4 reports the main split sizes, input dimensions, transformer architec-

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483 information about the statistical significance of the experiments?

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485 Justification: Tables 2 and 3 report bootstrap confidence intervals for AUC, Brier score, and

486 ECE. Section 5 states when paired bootstrap intervals include zero.

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